top of that base. "If you come to my office, you won't see CNC machines or PCB printing lines. There are already so many of them in this country," says Nekkanti. "The same manufacturing guys who build for aerospace also build for defence, industrial, everything. All the ingredients required for a thriving space ecosystem already existed."

But the base only helped if one knew what to ask it for. "For somebody who's been there, done that before, it's relatively fast," says Nekkanti. "When you don't know what you want, that's the problem."

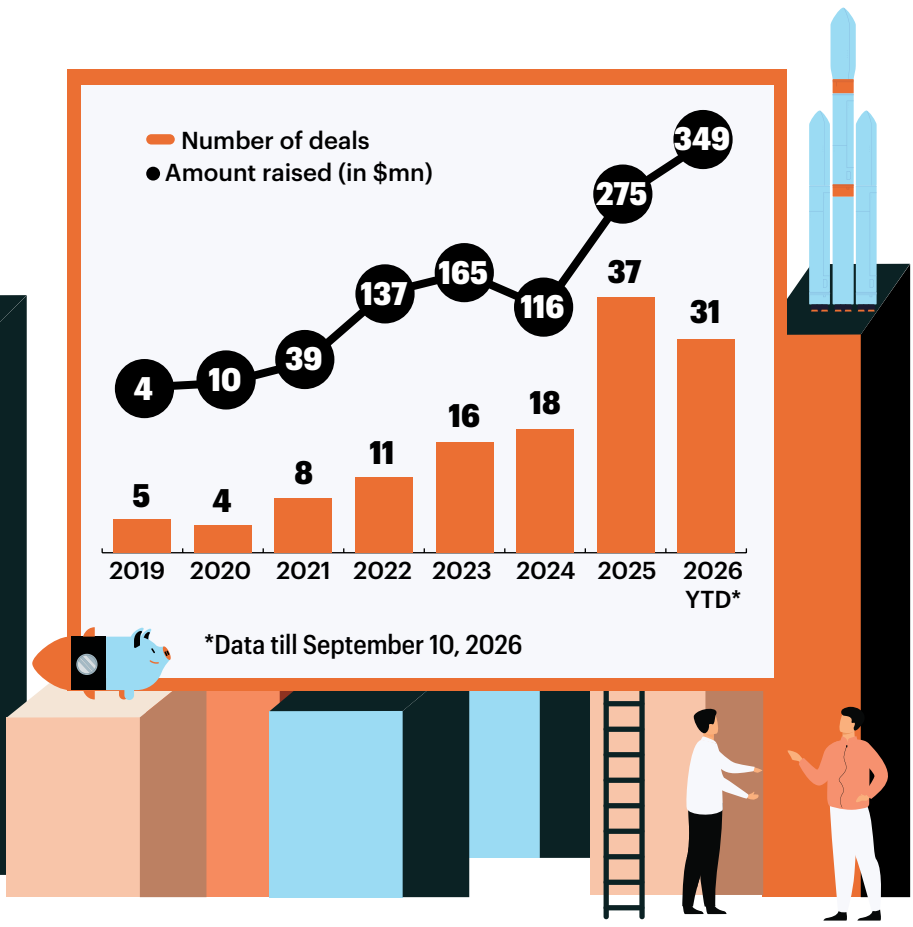
Beyond the manufacturing base, there were people. Several Isro engineers, including Skyroot Co-founders Chandana and Naga Bharath Daka, became entrepreneurs. Many other retired engineers became advisors to these new-age companies. Some scientists even joined them full-time. Being coached by the scientists who had worked under A.P.J. Abdul Kalam, when Isro was young and short of resources, was invaluable, according to Agnikul's Ravichandran.

"That generation of people were inspired. Isro was like a start-up then," says Ravichandran. "That scrappiness, in terms of getting together whatever it takes to build something, fire the engine—that's what I learned."

Isro's name did the rest. When Bengaluru-based Pixxel began selling hyperspectral data to customers in the US, for example, Isro's legacy did some of the early persuading.

"Isro thankfully had set that foundation and set up that reputation," says Awais Ahmed, Co-founder and CEO of Pixxel. "As a private company, we were still unproven, but once they saw that we were coming from India and that we had retired Isro scientists as advisors and some Isro scientists who had joined us full-time, we were taken quite seriously." The deepest inheritance is the know-how. Reaching orbit is a capability only a handful of countries hold, and India already had it. "We know what it means to build stuff that works in space. We know how to go to orbit," says Ravichandran. India was not starting at the beginning.

But the companies still could not own what they made. That changed in 2020, when private players finally got a policy push.



THE COMING-OF-AGE MOMENT

In May 2020, in the middle of the Covid-19 pandemic, Finance Minister Nirmala Sitharaman announced a series of stimulus measures as part of the government's Aatmanirbhar Bharat package. Among them was the creation of the Indian National Space Promotion and Authorisation Centre (IN-SPACe) to let private firms use the assets and institutional knowledge of Isro and the Department of Space, and participate in space activities.

Six weeks later, the Union cabinet, chaired by Prime Minister Narendra Modi, approved "far-